

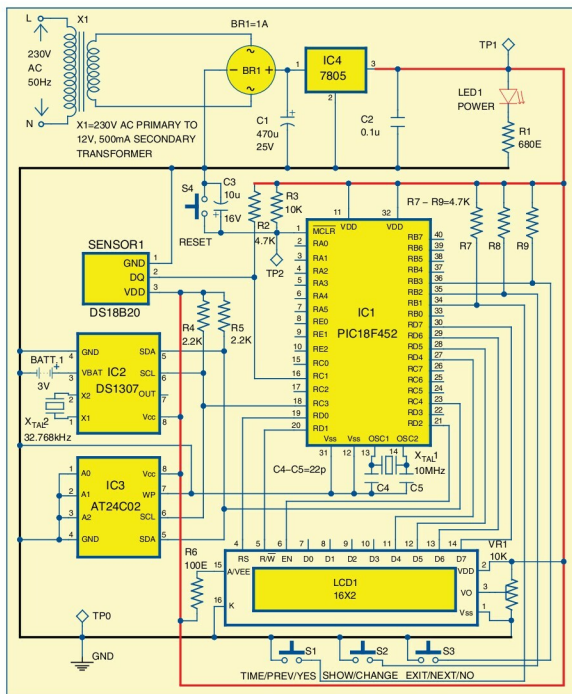


Fig. 2: Circuit diagram of real-time clock with temperature logger

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when S4 is pressed

supply when Vcc is not present. The SCL and SDA pins of DS1307 are interfaced to port pins RC3 and RC4 of microcontroller IC1 through pull-up resistors R4 and R5, respectively.

EEPROM. The minimum and maximum temperature of each day is

stored in serial EEPROM AT24C02. It provides 2048 bits of serial electrically erasable and programmable read-only memory (EEPROM) organised as 256 words of 8 bits each. The device is optimised for use in many industrial and commercial applications where low-power and low-voltage operation is essential. The microcontroller IC1 logs maximum and minimum temperatures for each day in EEPROM. If the EEPROM gets full, it shifts the contents of EEPROM by five positions to log new data. Address pins A0, A1, A2 of EEPROM AT24C02 are connected

digital thermometer SENSOR1 and shows them on the LCD. It also logs the maximum and minimum temperatures of the day in twin-wire serial EEPROM IC3. Port pins RD4 though RD7 of IC1 are connected to data pins D4 though D7 of LCD1, respectively. Port pins RD0 through RD2 of IC1 are connected to control pins RS, R/W and EN of LCD1. Multifunction switches S1 through S3 are connected to port pins RB1 through RB3 of IC1.

Software

The software program is written in 'C'

to ground and the serial communication pins SCL and SDA are interfaced to RC3 and RC4 of microcontroller IC1.

Microcontroller. PIC18F452 is the brain of the complete system. It is a powerful 10 MIPS CMOS FLASH-based 8-bit microcontroller and is upwards compatible with the PIC16C5X, PIC12CXXX, PIC16CXX and PIC17CXX devices, thus providing a seamless migration path of software code to higher levels of hardware integration. The PIC18F452 features a 'C' compiler-friendly development environment, 256 bytes of EEPROM, self-programming, an ICD, two capture/compare/PWM functions, eight channels of 10-bit analogue-to-digital (A/D) converter. Its synchronous serial port can be configured as either 3-wire serial peripheral interface or the 2-wire inter-integrated circuit bus and addressable universal asynchronous receiver transmitter.

The microcontroller continuously reads date and time from I²C-based real-time clock IC2, temperature from single-wire